

Purification of KRAS Fast-Exchange Mutants for Nucleotide Exchange and SOS-Binding Assays to Characterize Mutant KRAS-SOS Interactions

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Abstract

Small GTPases are a family of proteins responsible for cell signaling and growth in humans. They function as molecular switches, existing in the “on” (GTP-bound) or “off” (GDP-bound) states. KRAS is a small GTPase that regulates the MAPK signaling pathway responsible for cell growth, and mutation of KRAS is implicated in ~10% of all human cancers. KRAS is normally activated by the guanine nucleotide exchange factors (GEF), such as SOS, and is deactivated by GTPase activating proteins (GAP). The uncontrolled upregulation of KRAS activation via mutation leads to uncontrolled cell growth. Mutations can activate KRAS through enhanced nucleotide exchange, decreased GTP hydrolysis, or a combination of both. While direct targeting of KRAS in the clinic has achieved some success, tumors often develop resistance. In response, vertical inhibition strategies targeting the MAPK signaling pathway at multiple nodes (e.g. KRAS and SOS1) have been developed and are currently being explored by the field. We are interested in deciphering the underlying mechanism(s) of mutation-enhanced nucleotide exchange in KRAS, and how it relates to the SOS-catalyzed mechanism of nucleotide exchange. Here, we examine how different fast-exchange mutants of KRAS bind to SOS and influence the kinetics of nucleotide exchange using fluorescent nucleotide analogues and pull-down assays. These studies will help clarify the ways in which SOS1 works with mutant KRAS in the cell to enhance tumor growth, and ultimately, help predict whether vertical inhibition of SOS and KRAS is a rationally viable approach for different mutants of KRAS.

Objectives

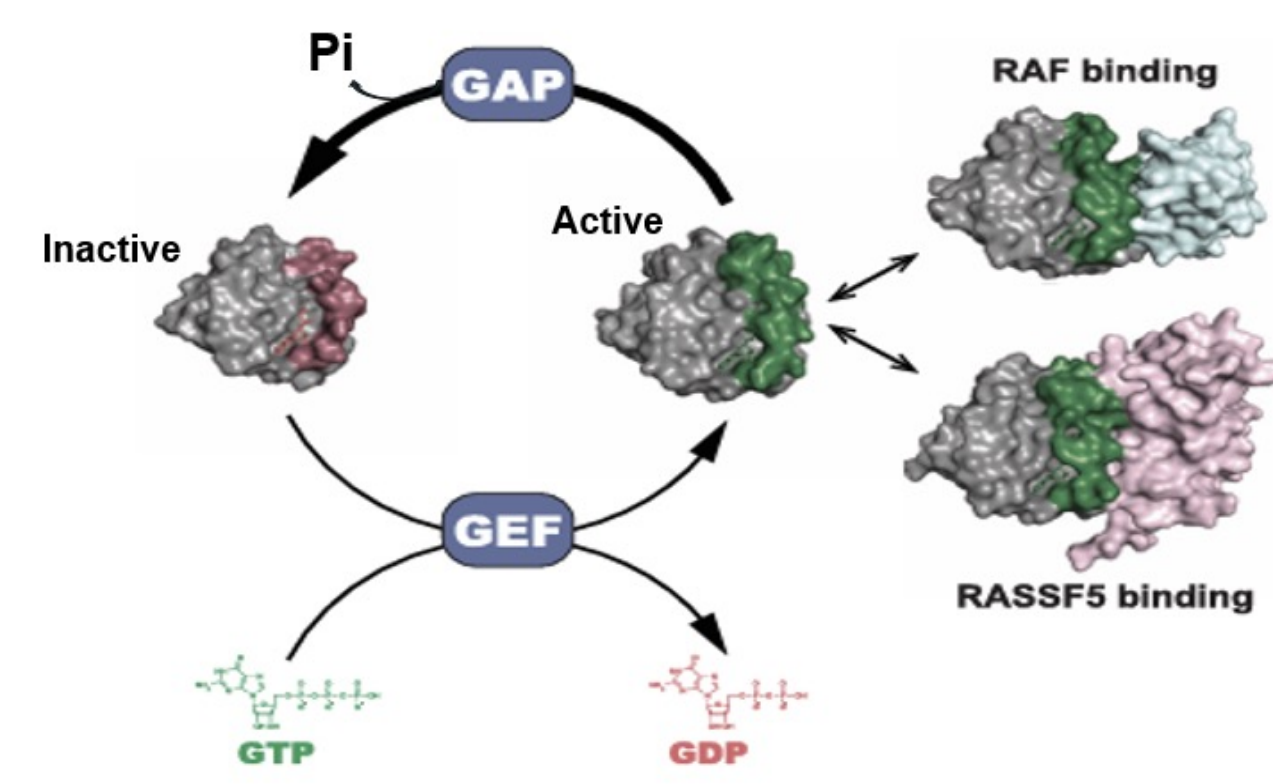


Figure 1. The canonical signaling cycle of RAS. When RAS is active, it can interact with downstream effectors like RAF and RASSF5, and propagate signaling; Johnson, C. W. et al., *Mol. Cell* **2022**, 82 (5), 950-968.

The aim of our experiments was to test if specific fast-exchange mutants of KRAS bind to SOS with a stronger affinity than others to determine if these mutants would cooperate with GEFs to enhance their active states (**Figure 1**). We hypothesize that some mutants with faster nucleotide exchange rates will bind more strongly to SOS, whereas others will not, due to differences in *how those mutants* enhance nucleotide exchange (**Figure 2**). For the experiments here, we predict that G13D will bind more strongly to SOS because the mutation induces active site changes similar to when wildtype KRAS binds SOS (**Figure 3**). However, fast-exchange mutants (e.g. KRAS A59E) that don't promote active site changes like G13D will bind weakly to SOS (**Figure 3**).

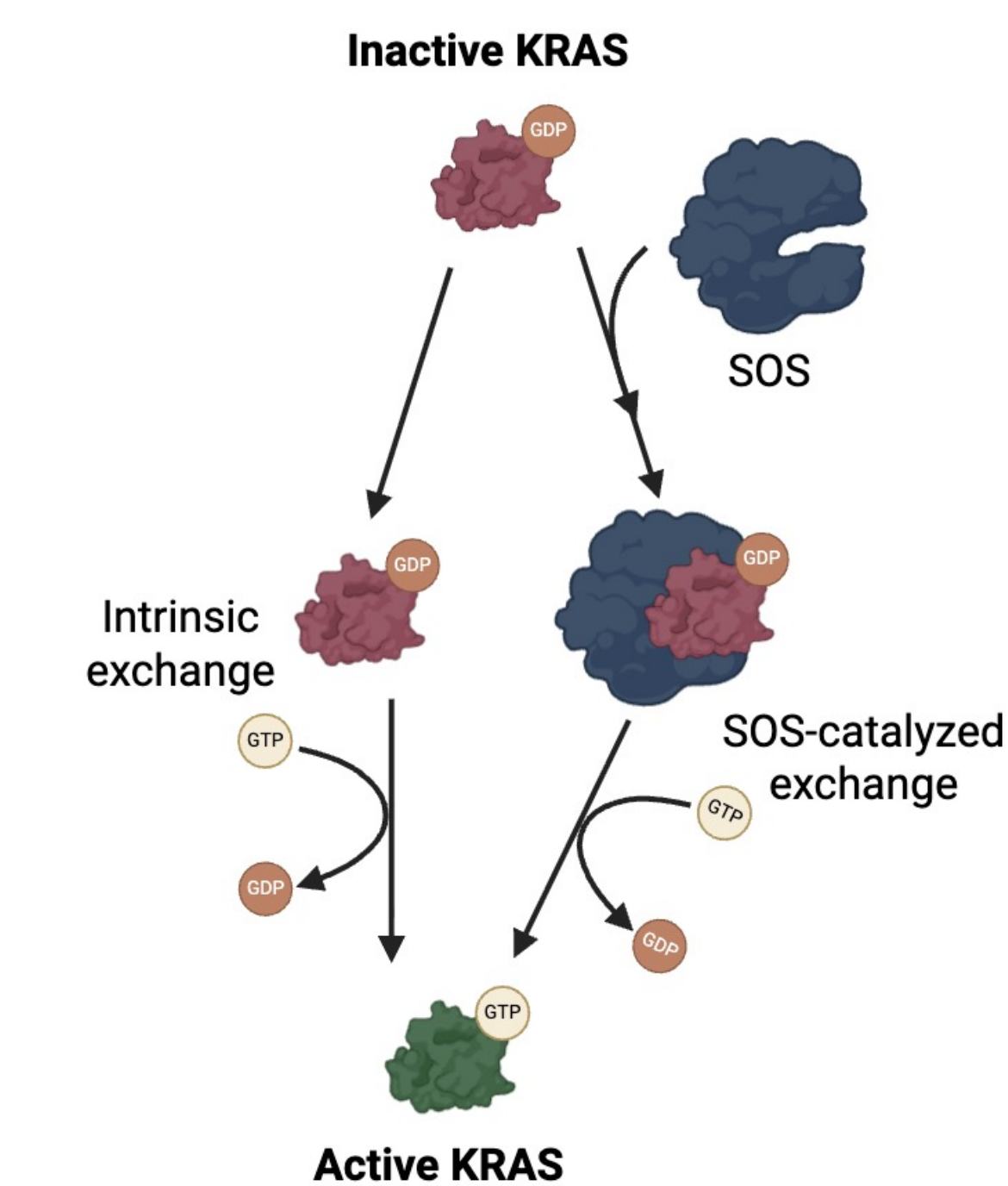


Figure 2. Intrinsic and SOS-catalyzed activation of KRAS. Catalytic mimicry hypothesis.

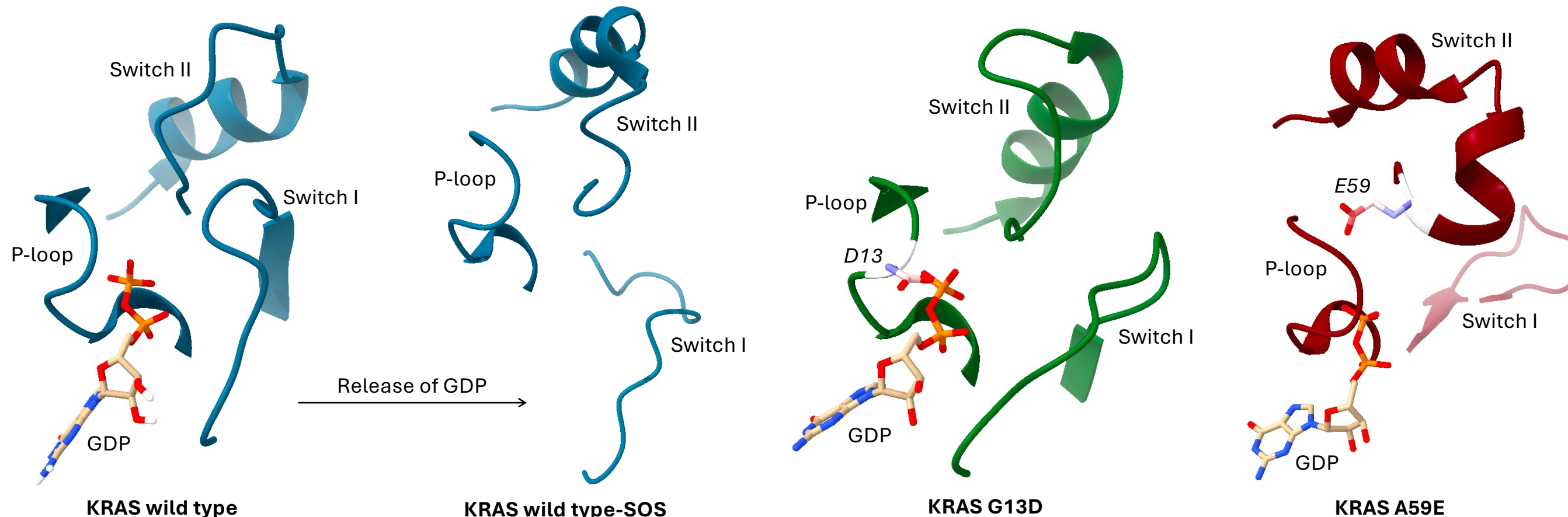


Figure 3. Structures of KRAS wild type, SOS-bound KRAS wild type (SOS hidden), G13D, and A59E effector lobes in their GDP-bound states. Shown are the P-loop (residues 10-17), switch I (residues 30-40), and switch II (residues 58-76)

Methods

KRAS and SOS were purified by sonication lysis and immobilized metal ion affinity pulldown (**Figure 4**). Protein expression constructs include a 6x His affinity tag or a cleavable 6HMCK affinity tag. KRAS exchange of mant-GDP for GTP was measured as the change in emission @ 440 nm; as mant-GDP is released by KRAS, its fluorescence decreases as compared to its KRAS-bound state (**Figure 5**). Binding by KRAS mutants to SOS was measured by Western blotting with RAS10 antibody. Incubation for binding assays was performed at both 4° C and 37° C (**Figure 6**).

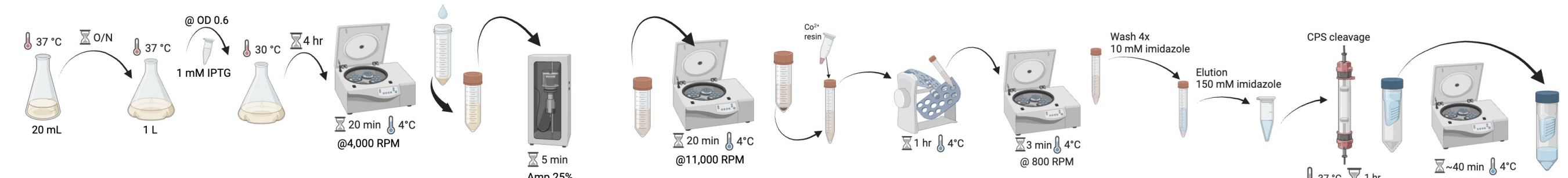


Figure 4. Immobilized metal ion affinity protein purification workflow.

